

EXECUTIVE SUMMARY

CFAR Agentic AI Hackathon

The CFAR Team Building Agentic AI Hackathon provides a hands-on opportunity for researchers and engineers to collaboratively explore how agentic AI can augment the practice of research itself. Teams will build multi-agent systems with autonomous or semi-autonomous workflows that take on real tasks from CFAR's daily research activities, such as proposal development, paper reviewing, literature survey, and conference planning, using reasoning, tool usage, and dynamic planning.

The primary goal is to move beyond conventional single-prompt chatbots, which cannot handle such tasks reliably, and to develop sophisticated agent architectures (e.g., planner-executor, generator-verifier, or human-in-the-loop systems) that can support research activities across CFAR and IAIC beyond the hackathon.

Judging Criteria

Projects will be evaluated on the criteria below by a judging panel comprising Dr Su Yi, Dr Lou Jing, Prof Ong Yew Soon and Prof Ivor Tsang.

- **Innovativeness (25%):** Judges will evaluate the novelty and creativity of the proposed agent architecture or workflow.
- **Teamwork (20%):** Teams must demonstrate effective collaboration and provide evidence that all 5–6 members contributed to the final solution.
- **Reusable Asset Potential (20%):** Solutions will be scored heavily on the likelihood that the prototype can be extended into a useful, permanent CFAR research or productivity capability.
- **Technical Rigor (20%):** Evaluates the quality of reasoning, tool integration, verification, and overall engineering.
- **Agentic Design (10%):** Assesses whether the agent's behavior genuinely adds value compared to a simple, fixed LLM workflow.
- **Demonstration/Evaluation (5%):** Requires clear evidence that the approach actually works through meaningful examples.

Hackathon Challenge Tracks

The four challenges below each target one stage of the research lifecycle: securing funding (P1), assessing research quality (P2), understanding the field (P3), and disseminating the work (P4). Teams can choose one of them or propose their own.

- **P1. Proposal Generation Agent:** A system that analyzes grant calls, identifies research gaps, evaluates competitors and potential collaborators, and budgets resources to craft competitive proposal strategies.
- **P2. Scientific Paper Reviewer:** A structured workflow utilizing specialized agents (e.g., Novelty, Methodology, Reproducibility) to rigorously critique research papers.
- **P3. Literature Survey Agent:** An autonomous system that retrieves papers, builds taxonomies, traces citations, and identifies emerging research gaps.
- **P4. Travel Planning Agent:** A dynamic agent that balances flights, budgets, conference schedules, and institutional policies, adjusting instantly to schedule changes or cancellations.
- **Custom Project:** Teams may propose their own complex task, provided it demonstrates meaningful agentic behavior.

Timeline & Deliverables

Teams are encouraged to divide responsibilities based on the requirements of their proposed solution rather than their individual research disciplines, with one nominated coordinator per team.

ACTIVITY	TIMELINE
Challenge Announcement	27 Aug 2026
Team formation	By 31 Aug 2026
Development Period	01 Sep to 10 Sep 2026
Mandatory Slide Submission	10 Sep 2026 12:00
Presentation	11 Sep 2026
Judging & Results announcement	11 Sep 2026

Mandatory Deliverable: A short slide deck covering the problem, architecture, tools used, key design decisions, evaluation evidence, and the path toward a reusable CFAR asset.

Optional (Highly Encouraged): A functioning proof-of-concept prototype (preferably a Docker container) demonstrating key agent interactions, prioritizing technical ideas over UI design.

CFAR TEAM BUILDING

AGENTIC AI HACKATHON

A hands-on experience to team-build innovative and reusable Agentic AI systems

The hackathon is not a “best demo” contest.

Teams are expected to show a clear problem, a genuinely agentic solution, strong collaboration, and a prototype that could continue to create value for CFAR after the event.

1. Purpose and Success Definition

The CFAR Team Building Agentic AI Hackathon gives researchers and engineers a *focused, collaborative* opportunity to design and implement Agentic AI workflows for practical research or operational challenges.

The goal is to move beyond a single prompt or chatbot and explore systems that can reason, plan, use tools, delegate work, verify results, adapt to new information and involve humans where appropriate:

- Multi-agent collaboration and delegation
- Planning and task decomposition
- Tool, API and information retrieval
- Reasoning over heterogeneous information
- Verification, critique and uncertainty handling
- Memory, state management and dynamic replanning
- Human-in-the-loop interaction and responsible behaviour
- Objective evaluation of agent performance

2. Team Formation and Working Style

Teams will normally consist of 5–6 members. A team coordinator should keep the team aligned, but contributions should include, but not limited to the following:

- Form around the problem and workflow—not around existing disciplines or job titles.
- Assign clear roles, while making sure the solution reflects integrated contributions rather than disconnected sub-components.
- Make collaboration visible: explain who contributed what, how interfaces were agreed, and how the team resolved design trade-offs.
- Use the hackathon to learn together; interesting, failed experiments and lessons are valuable evidence of teamwork.

3. Timeline

Activity	Date / Duration
Team formation announced	27 Aug 2026
Challenge announced	31 Aug 2026
Development period	01 to 10 Sep 2026
Slides due	10 Sep 2026 12:00
Team presentations + demonstrations	11 Sep 2026
Presentation time	Approx 10 minutes per team
Results	Following evaluation

4. Challenge

Each team must design an Agentic AI workflow that autonomously or semi-autonomously solves a complex real-world task involving reasoning, planning, information gathering and/or tool use.

Choose one of the proposed projects below; or

Propose your own Agentic AI workflow, provided it demonstrates meaningful agentic behaviour rather than primarily conventional LLM generation.

There is no requirement to use the same model, framework, programming language or architecture. Explain why your design is appropriate for the problem.

P1. Research Proposal Strategy Agent

Problem: Help a research team turn a funding call and available CFAR expertise/resources into a stronger proposal strategy and draft.

Possible agents: Grant analysis; research landscape; prior-award and competitor analysis; team formation; technical architecture; budget/resources; proposal critique; proposal coordination.

Desired outputs / key challenge: Outputs may include research gaps, differentiation, team recommendations, work packages, risks, resources, impact and a final proposal outline. Key issue: separate evidence, assumptions and speculation.

P2. Scientific Paper Critic and Reviewer Agent

Problem: Conduct a structured, rigorous review rather than simply asking an LLM to “review this paper.”

Possible agents: Paper understanding; novelty; methodology; experiments; statistics; reproducibility; adversarial review; author response; area-chair synthesis.

Desired outputs / key challenge: Outputs may include strengths, weaknesses, questions, correctness/novelty/experimental assessments, confidence and evidence-backed recommendation. Advanced ideas: reviewer disagreement, debate, citation/math/code checking.

P3. Autonomous Literature Survey and Research Landscape Agent

Problem: Build a research landscape that goes beyond summarising papers and instead reveals structure, evolution, contradictions and gaps.

Possible agents: Search strategy; retrieval; relevance; paper analysis; taxonomy; citation lineage; trends; contradictions; research gaps; survey editing.

Desired outputs / key challenge: Outputs may include taxonomy, chronology, comparisons, key papers/groups, datasets, conflicting findings, trends, open problems and research opportunities. Verify references and trace claims to sources.

P4. Intelligent Conference Travel Planning Agent

Problem: Create and dynamically update a conference travel plan under multiple constraints.

Possible agents: Conference programme; flights; accommodation; transport; scheduling; networking; budget; policy; risk/contingency; final coordination.

Desired outputs / key challenge: The workflow should respond to changes such as cancellation, schedule changes, new meetings, hotel unavailability or reduced budget without unnecessarily recomputing everything.

5. Teams Proposing Their Own Project

Alternative ideas are encouraged. A proposed project should be selected because agentic behaviour creates a meaningful advantage—not simply because an LLM can generate a plausible answer.

6. Technical Scope

Teams are free to choose the technical approach. Candidate building blocks include agent orchestration frameworks, LLM APIs or open models, retrieval/search, databases, optimisation, computer vision and other relevant components.

Design questions judges may probe:

- What should be decided by an LLM, and what should be deterministic or algorithmic?
- When should an agent call a tool or delegate to another agent?
- How does the system verify outputs and detect weak evidence?
- How is state shared, and how are conflicting recommendations resolved?
- Where should humans remain in the decision loop?
- How does the workflow recover from failure?
- How is performance evaluated objectively?

7. Deliverables and Presentation

Mandatory: a short slide deck explaining the solution. A working prototype is strongly encouraged but does not need to be production-grade.

- Problem and motivation
- Agentic architecture and agent roles
- Tools / external resources
- Workflow and orchestration
- Key technical decisions
- Demonstration or example execution
- Evaluation and evidence
- Limitations and failure modes
- Path towards a reusable CFAR asset

Suggested 10-minute format

2 min — problem and why an agentic workflow is appropriate

3 min — architecture, agents and orchestration

3 min — demonstration

1 min — evaluation, failures and lessons learned

1 min — reusable-asset potential and next steps

8. Judging Criteria — What Wins

The panel will score the solution as a whole. The weighting below places stronger emphasis on the three outcomes the hackathon is intended to create: innovation, team collaboration and reusable CFAR value.

Criterion	Weight	Judging focus	What strong evidence looks like
Innovation	25%	Novelty and creativity of the problem framing, agent architecture or workflow.	A clear “why this is different” story; thoughtful use of agents/tools; a design choice that creates new capability or a better way of working.
Teamwork	20%	Quality of collaboration and integration across team members.	Complementary contributions are visible; interfaces fit together; trade-offs were resolved collectively; the final workflow reflects team synthesis.
Reusable CFAR Asset	20%	Likelihood that the prototype can become a useful CFAR capability.	A working core that another team could extend; reusable components, prompts, tools or evaluation assets; clear next-step path and realistic adoption potential.
Technical Rigor	20%	Architecture quality, reasoning, tool integration, verification and engineering discipline.	Sound system design; sensible division of LLM vs deterministic logic; robust tool use; failure handling; evidence-based evaluation.
Agentic Design	10%	Whether agent behaviour genuinely adds value over a simple prompt or fixed workflow.	Runtime decisions, delegation, replanning, verification, uncertainty handling and adaptation are visible in the demo.
Demonstration & Evidence	5%	Clarity of proof that the approach works.	A focused demo, meaningful test cases, measurable or inspectable evidence, and candid discussion of failures and limits.

Total: 100%

9. What Constitutes a Strong Agentic AI Solution?

Strong projects demonstrate that the system can make meaningful decisions about what to do next rather than executing only a predetermined sequence of prompts.

For example, a stronger workflow might:

Observe → Reason → Delegate → Use Tool → Evaluate Result → Critique → Re-plan → Act

rather than:

Prompt 1 → Prompt 2 → Prompt 3 → Final Answer

Teams are encouraged to demonstrate behaviours such as:

- Deciding which specialist agent should handle a task
- Selecting tools based on the current situation
- Revising a plan after receiving new information
- Identifying insufficient evidence and searching for more
- Detecting disagreement between agents
- Requesting human confirmation for important decisions
- Critiquing and revising an intermediate result
- Recovering from unsuccessful tool calls
- Explicitly estimating uncertainty or confidence

10. Additional Recognition and Expected Outcomes

Optional recognition may be given for:

- Best Agentic AI Architecture
- Most Innovative Solution
- Best Technical Implementation
- Best Demonstration
- Best Potential Reusable CFAR Asset
- Most Interesting Failure / Lesson Learned

The final category encourages teams to share unexpected behaviour and useful technical lessons rather than hiding unsuccessful experiments.

By the end of the hackathon, CFAR should have greater cross-disciplinary interaction, practical experience with agentic systems, a collection of prototypes, and a stronger pipeline of ideas that can evolve into research, engineering or reusable internal assets.

Final message to teams

Build something that is innovative enough to stand out, collaborative enough to show the power of the team, and reusable enough to matter after the hackathon ends.